Estimating proportions: 1 or 2 samples

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Point estimator of P in a binomial experiment
Where X = # successes in n trials

$$\hat{p} = \frac{X}{n}$$

$X_1, X_2, \ldots X_n$

$$\hat{p} = \frac{\sum X_i}{n} = \bar{X}$$

$$\hat{p}$$

$$\mu_{\hat{p}} = E[\hat{p}] = \frac{E(X)}{n} = \frac{np}{n} = p$$

$$\sigma^2_{\hat{p}} = V(\hat{p}) = \frac{V(X)}{n^2} = \frac{npq}{n^2} = \frac{pq}{n}$$

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}} \sim N(0,1)$$

$$P\left(-z_{\alpha/2} < \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}} < z_{\alpha/2}\right) = 1 - \alpha$$

$$P\left(-z_{\alpha/2}\sqrt{\frac{pq}{n}} < \hat{p} - p < z_{\alpha/2}\sqrt{\frac{pq}{n}}\right) = 1 - \alpha$$

$$P\left(\hat{p} - z_{\alpha/2}\sqrt{\frac{pq}{n}} < p < \hat{p} + z_{\alpha/2}\sqrt{\frac{pq}{n}}\right) = 1 - \alpha$$

Any problem?

x, n, calculate the CI?

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{pq}{n}} \quad ; \quad q = 1 - p$$

Large-Sample Confidence Intervals for p

If $\hat{p}$ is the proportion of successes in a random sample of size n and $\hat{q} = 1 - \hat{p}$, an approximate $100(1-\alpha)\%$ confidence interval, for the binomial parameter p is given by (method 1)

$$\hat{p} - z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}} < p < \hat{p} + z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}$$

used when
$n\hat{p} \geq 5$ $\boxed{\text{AND}}$
$n(1-\hat{p}) \geq 5$

or by (method 2)

more accurate but difficult

$$\frac{\hat{p} + \frac{z^2_{\alpha/2}}{2n}}{1 + \frac{z^2_{\alpha/2}}{n}} - \frac{z_{\alpha/2}}{1 + \frac{z^2_{\alpha/2}}{n}}\sqrt{\frac{\hat{p}\hat{q}}{n} + \frac{z^2_{\alpha/2}}{4n^2}} < p < \frac{\hat{p} + \frac{z^2_{\alpha/2}}{2n}}{1 + \frac{z^2_{\alpha/2}}{n}} + \frac{z_{\alpha/2}}{1 + \frac{z^2_{\alpha/2}}{n}}\sqrt{\frac{\hat{p}\hat{q}}{n} + \frac{z^2_{\alpha/2}}{4n^2}},$$

where $z_{\alpha/2}$ is the z-value leaving an area of $\alpha/2$ to the right.

Example 9.14: In a random sample of $n = 500$ families owning television sets in the city of Hamilton, Canada, it is found that $x = 340$ subscribe to HBO. Find a 95% confidence interval for the actual proportion of families with television sets in this city that subscribe to HBO.

$z_{0.025} = 1.96$

$$\hat{p} = \frac{340}{500} = 0.68, \quad \hat{q} = 0.32$$

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}} = (0.639, 0.721) \quad \text{Method 1}$$

$0.6391 < p < 0.7209.$

If we use method 2, we can obtain

$$\frac{0.68 + \frac{1.96^2}{(2)(500)}}{1 + \frac{1.96^2}{500}} \pm \frac{1.96}{1 + \frac{1.96^2}{500}}\sqrt{\frac{(0.68)(0.32)}{500} + \frac{1.96^2}{(4)(500^2)}} = 0.6786 \pm 0.0408,$$

$$(0.6378, 0.719) \quad \text{Method 2}$$

Theorem 9.3: If $\hat{p}$ is used as an estimate of p, we can be $100(1-\alpha)\%$ confident that the error will not exceed $z_{\alpha/2}\sqrt{\hat{p}\hat{q}/n}$.

$$\boxed{e \leq z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}}$$

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Theorem 9.4: If $\hat{p}$ is used as an estimate of p, we can be $100(1-\alpha)\%$ confident that the error will be less than a specified amount e when the sample size is approximately

$$\boxed{n = \frac{z_{\alpha/2}^2 \hat{p}\hat{q}}{e^2}.}$$

n can only be computed with knowledge of $\hat{p}$??

Example 9.15: How large a sample is required if we want to be 95% confident that our estimate of p in Example 9.14 is within 0.02 of the true value?

$$n = \frac{z_{\alpha/2}^2 \hat{p}\hat{q}}{e^2} = 2090$$

$z_{\alpha/2} = 1.96$, $e = 0.02$, $\hat{p} = 0.68$

If we do not know p, then we can consider the extreme case

$f = p(1-p)$ will be maximized when ?

$f' = 1 - 2p = 0 \Rightarrow p = 1/2$

$f'' = -2 < 0$

Q Can we use this bound even if we have some idea about $\hat{p}$?
sure, but

$\Rightarrow$ degree of CI increases

$\hat{p} = 0.68$; $n = 500$
$e = 0.02$; $\alpha = 0.05$

2090 vs 2401

Theorem 9.5: If $\hat{p}$ is used as an estimate of p, we can be at least $100(1-\alpha)\%$ confident that the error will not exceed a specified amount e when the sample size is

$$\left(n = \frac{z_{\alpha/2}^2 \hat{p}\hat{q}}{e^2}\right) \text{ OR } \boxed{n = \frac{z_{\alpha/2}^2}{4e^2}.}$$

Example 9.16: How large a sample is required if we want to be at least 95% confident that our estimate of p in Example 9.14 is within 0.02 of the true value?

$z_{\alpha/2} = 1.96$, $e = 0.02$

$n \simeq 2401$

[redacted]

The two populations are independent

Large-Sample Confidence Interval for $p_1 - p_2$

If $\hat{p}_1$ and $\hat{p}_2$ are the proportions of successes in random samples of sizes n_1 and n_2, respectively, $\hat{q}_1 = 1 - \hat{p}_1$, and $\hat{q}_2 = 1 - \hat{p}_2$, an approximate $100(1-\alpha)\%$ confidence interval for the difference of two binomial parameters, $p_1 - p_2$, is given by

$V(\hat{p}_1) + V(\hat{p}_2)$

$$(\hat{p}_1 - \hat{p}_2) - z_{\alpha/2}\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}} < p_1 - p_2 < (\hat{p}_1 - \hat{p}_2) + z_{\alpha/2}\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}},$$

where $z_{\alpha/2}$ is the z-value leaving an area of $\alpha/2$ to the right.

Example 9.17: A certain change in a process for manufacturing component parts is being considered. Samples are taken under both the existing and the new process so as where $z_{\alpha/2}$ is the z-value leaving an area of $\alpha/2$ to the right.

Example 9.17: A certain change in a process for manufacturing component parts is being considered. Samples are taken under both the existing and the new process so as to determine if the new process results in an improvement. If 75 of 1500 items from the existing process are found to be defective and 80 of 2000 items from the new process are found to be defective, find a 90% confidence interval for the true difference in the proportion of defectives between the existing and the new process.

$z_{0.05} = 1.645$

- proportion of XYZ [p= counts / n]

X : # of successes in n trials

$\sim Bin(n, p) \Rightarrow E(X) = np \; ; \; V(X) = npq$

# of accidents $\quad q = 1-p$

$\Rightarrow$ accident rate at GC in a month

$\boxed{\hat{p}} = \frac{X}{n} \qquad X \simeq \underline{\sum_{i=1}^{n} Y_i} \quad ; \; Y_i \sim Bern(0,1)$

$\hat{p} \rightarrow p \qquad \frac{X}{n} \approx \bar{Y}$

$E[\hat{p}] = E\left[\frac{X}{n}\right] = \frac{np}{n} = p$

$V(\hat{p}) = V\left(\frac{X}{n}\right) = \frac{1}{n^2} npq = \frac{pq}{n}$

$$\frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}} \sim N(0,1) \rightarrow CLT$$

$X_i \sim N(\mu, \sigma^2)$; σ^2 unknown

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$$

$$P\left(\chi^2_{1-\alpha/2} < \chi^2 < \chi^2_{\alpha/2}\right) = 1-\alpha$$

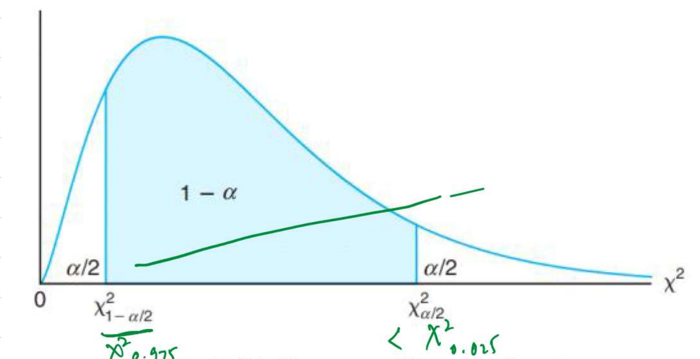

Figure 9.7: $P(\chi^2_{1-\alpha/2} < X^2 < \chi^2_{\alpha/2}) = 1 - \alpha$.

$$P\left(\chi^2_{1-\alpha/2} < \frac{(n-1)s^2}{\sigma^2} < \chi^2_{\alpha/2}\right) = 1-\alpha$$

$$P\left(\frac{(n-1)s^2}{\chi^2_{\alpha/2}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}\right) = 1-\alpha$$

Confidence Interval for σ^2 If s^2 is the variance of a random sample of size n from a normal population, a $100(1-\alpha)\%$ confidence interval for σ^2 is

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}},$$

where $\chi^2_{\alpha/2}$ and $\chi^2_{1-\alpha/2}$ are χ^2-values with $v = n-1$ degrees of freedom, leaving areas of $\alpha/2$ and $1-\alpha/2$, respectively, to the right.

Example 9.18: The following are the weights, in decagrams, of 10 packages of grass seed distributed by a certain company: 46.4, 46.1, 45.8, 47.0, 46.1, 45.9, 45.8, 46.9, 45.2, and 46.0. Find a 95% confidence interval for the variance of the weights of all such packages of grass seed distributed by this company, assuming a normal population.

0.135
0.953

$$s^2 = \frac{1}{9}\sum(x_i - \bar{x})^2 = 0.286 \ ?$$

$\nu = 9 \Rightarrow \chi^2_{0.025} = 19.023$, and $\chi^2_{0.975} = 2.700$

$\nu = 9 \Rightarrow \chi^2_{0.025} = 19.023$, and $\chi^2_{0.975} = 2.700$

$\sigma_1^2 - \sigma_2^2 = 0$, $\neq 0$ or $\frac{\sigma_1^2}{\sigma_2^2} = 1$, $\neq 1$

- Variances are always ≥ 0
- Unit if x_i are in meters then ?

$\frac{(n_1-1)s_1^2}{\sigma_1^2} \sim \chi^2_{n_1-1}$

independent

$\frac{(n_2-1)s_2^2}{\sigma_2^2} \sim \chi^2_{n_2-1}$

$\frac{\chi^2_{v_1}/v_1}{\chi^2_{v_2}/v_2} \sim F_{v_1, v_2}$

$$F = \frac{\frac{\cancel{(n_1-1)} s_1^2}{\sigma_1^2 \cancel{(n_1-1)}}}{\frac{\cancel{(n_2-1)} s_2^2}{\sigma_2^2 \cancel{(n_2-1)}}} \sim F_{n_1-1, n_2-1}$$

$$= \frac{\sigma_2^2 s_1^2}{\sigma_1^2 s_2^2}$$

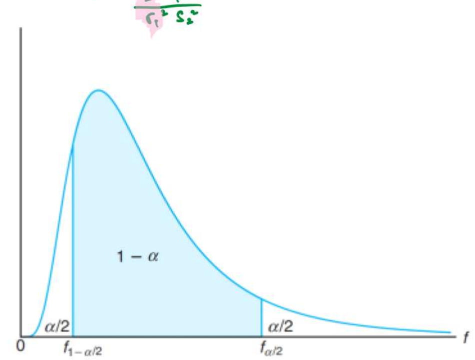

Figure 9.8: $P[f_{1-\alpha/2}(v_1, v_2) < F < f_{\alpha/2}(v_1, v_2)] = 1 - \alpha$.

$$P\left(f_{1-\alpha/2}(v_1, v_2) < \frac{\sigma_2^2 s_1^2}{\sigma_1^2 s_2^2} < f_{\alpha/2}(v_1, v_2)\right) = 1-\alpha$$

$$P\left(\frac{1}{f_{\alpha/2}(v_1, v_2)}\left(\frac{s_1^2}{s_2^2}\right) < \frac{\sigma_1^2}{\sigma_2} < \frac{1}{f_{1-\alpha/2}(v_1, v_2)}\left(\frac{s_1^2}{s_2^2}\right)\right) = 1-\alpha$$

$$\boxed{F_{\alpha/2, v_1, v_2} = \frac{1}{F_{1-\alpha/2, v_2, v_1}}}$$

$$f_{1-\alpha/2}(v_1, v_2) = \frac{1}{f_{\alpha/2}(v_2, v_1)}$$

Confidence Interval for σ_1^2/σ_2^2

If s_1^2 and s_2^2 are the variances of independent samples of sizes n_1 and n_2, respectively, from normal populations, then a $100(1-\alpha)\%$ confidence interval for σ_1^2/σ_2^2 is

$$\frac{s_1^2}{s_2^2}\frac{1}{f_{\alpha/2}(v_1, v_2)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{s_1^2}{s_2^2} f_{\alpha/2}(v_2, v_1),$$

where $f_{\alpha/2}(v_1, v_2)$ is an f-value with $v_1 = n_1 - 1$ and $v_2 = n_2 - 1$ degrees of freedom, leaving an area of $\alpha/2$ to the right, and $f_{\alpha/2}(v_2, v_1)$ is a similar f-value with $v_2 = n_2 - 1$ and $v_1 = n_1 - 1$ degrees of freedom.

Example 9.19: A confidence interval for the difference in the mean orthophosphorus contents, measured in milligrams per liter, at two stations on the James River was constructed in Example 9.12 on page 290 by assuming the normal population variance to be unequal. Justify this assumption by constructing 98% confidence intervals for σ_1^2/σ_2^2 and for σ_1/σ_2, where σ_1^2 and σ_2^2 are the variances of the populations of orthophosphorus contents at station 1 and station 2, respectively.

[1.851 , 7.549]

$n_1 = 15$, $n_2 = 12$, $s_1 = 3.07$ and $s_2 = 0.80$, $\alpha = 0.02$

$f_{0.01}(14, 11) \approx 4.30$ and $f_{0.01}(11, 14) \approx 3.87$

Summary of all CIs

24 September 2025 08:38

Case	Assumption	Test Statistic	df / Notes	Confidence Intervals / Bounds
One-sample Z $\mu = \mu_0$	Normal population, known σ	$Z = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$	$Z \sim N(0,1)$. Exact under normality; CLT for large n.	For μ: Two-sided: $\bar{X} \pm z_{1-\alpha/2}\dfrac{\sigma}{\sqrt{n}}$ Lower: $\mu > \bar{X} - z_{1-\alpha}\dfrac{\sigma}{\sqrt{n}}$ Upper: $\mu < \bar{X} + z_{1-\alpha}\dfrac{\sigma}{\sqrt{n}}$
One-sample t $\mu = \mu_0$	Normal population, unknown σ	$t = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}}$	$\mathrm{df} = n - 1$.	For μ: Two-sided: $\bar{X} \pm t_{1-\alpha/2,n-1}\dfrac{S}{\sqrt{n}}$ Lower: $\mu > \bar{X} - t_{1-\alpha,n-1}\dfrac{S}{\sqrt{n}}$ Upper: $\mu < \bar{X} + t_{1-\alpha,n-1}\dfrac{S}{\sqrt{n}}$
Two-sample Z independent $\mu_1 = \mu_2$	Normal populations, known σ_1, σ_2	$Z = \dfrac{\bar{X}_1 - \bar{X}_2}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$	$Z \sim N(0,1)$. Independent samples. $SE = \dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}$	For $\Delta = \mu_1 - \mu_2$: Two-sided: $(\bar{X}_1 - \bar{X}_2) \pm z_{1-\alpha/2}SE$ Lower: $\Delta > (\bar{X}_1 - \bar{X}_2) - z_{1-\alpha}SE$ Upper: $\Delta < (\bar{X}_1 - \bar{X}_2) + z_{1-\alpha}SE$
Two-sample pooled t independent $\mu_1 = \mu_2$	Normal populations, unknown but equal variances	$t = \dfrac{\bar{X}_1 - \bar{X}_2}{S_p\sqrt{1/n_1 + 1/n_2}}$ $S_p^2 = \dfrac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1 + n_2 - 2}$	$\mathrm{df} = \nu = n_1 + n_2 - 2$. Requires equal variances. $SE = S_p\sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}$	For $\Delta = \mu_1 - \mu_2$: Two-sided: $(\bar{X}_1 - \bar{X}_2) \pm t_{1-\alpha/2,\nu}SE$ Lower: $\Delta > (\bar{X}_1 - \bar{X}_2) - t_{1-\alpha,\nu}SE$ Upper: $\Delta < (\bar{X}_1 - \bar{X}_2) + t_{1-\alpha,\nu}SE$
Welch's t independent $\mu_1 = \mu_2$	Normal populations, unknown and possibly unequal variances	$t = \dfrac{\bar{X}_1 - \bar{X}_2}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}$	$\nu \approx \dfrac{(S_1^2/n_1 + S_2^2/n_2)^2}{\frac{(S_1^2/n_1)^2}{(n_1-1)} + \frac{(S_2^2/n_2)^2}{(n_2-1)}}$ $SE = \dfrac{S_1^2}{n_1} + \dfrac{S_2^2}{n_2}$	For $\Delta = \mu_1 - \mu_2$: Two-sided: $(\bar{X}_1 - \bar{X}_2) \pm t_{1-\alpha/2,\nu}SE$ Lower: $\Delta > (\bar{X}_1 - \bar{X}_2) - t_{1-\alpha,\nu}SE$ Upper: $\Delta < (\bar{X}_1 - \bar{X}_2) + t_{1-\alpha,\nu}SE$

Case	Assumption	Test Statistic	df / Notes	Confidence Intervals / Bounds
Paired Z $\mu_D = 0$	Differences $D_i = X_i - Y_i$ normal, known σ_D	$Z = \dfrac{\bar{D}}{\sigma_D/\sqrt{n}}$	$Z \sim N(0,1)$. Usually paired t is used.	For μ_D: Two-sided: $\bar{D} \pm z_{1-\alpha/2}\dfrac{\sigma_D}{\sqrt{n}}$ Lower: $\mu_D > \bar{D} - z_{1-\alpha}\dfrac{\sigma_D}{\sqrt{n}}$ Upper: $\mu_D < \bar{D} + z_{1-\alpha}\dfrac{\sigma_D}{\sqrt{n}}$
Paired t $\mu_D = 0$	Differences $D_i = X_i - Y_i$ normal, unknown σ_D	$t = \dfrac{\bar{D}}{S_D/\sqrt{n}}$	df $= n - 1$. Equivalent to one-sample t on differences.	For μ_D: Two-sided: $\bar{D} \pm t_{1-\alpha/2,n-1}\dfrac{S_D}{\sqrt{n}}$ Lower: $\mu_D > \bar{D} - t_{1-\alpha,n-1}\dfrac{S_D}{\sqrt{n}}$ Upper: $\mu_D < \bar{D} + t_{1-\alpha,n-1}\dfrac{S_D}{\sqrt{n}}$
One-sample proportion $p = p_0$	Binomial/Bernoulli population; large-sample approximation	$Z = \dfrac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$	$Z \sim N(0,1)$. Requires sufficiently large expected counts.	For p: Two-sided: $\hat{p} \pm z_{1-\alpha/2}\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n}}$ Lower: $p > \hat{p} - z_{1-\alpha}\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n}}$ Upper: $p < \hat{p} + z_{1-\alpha}\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n}}$ *(Wald intervals)*
Two-sample proportions $p_1 = p_2$	Independent binomial populations; large-sample approximation	$Z = \dfrac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)}}$ $\hat{p} = \dfrac{x_1 + x_2}{n_1 + n_2}$	Pooled estimate used for testing $H_0 : p_1 = p_2$.	For $\Delta = p_1 - p_2$: Two-sided: $(\hat{p}_1 - \hat{p}_2) \pm z_{1-\alpha/2}SE$ Lower: $\Delta > (\hat{p}_1 - \hat{p}_2) - z_{1-\alpha}SE$ Upper: $\Delta < (\hat{p}_1 - \hat{p}_2) + z_{1-\alpha}SE$ where $SE = \sqrt{\dfrac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \dfrac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$ *(unpooled SE for CI)*

Hypothesis Tests and Confidence Intervals

Tests for Population Variances

Case	Assumption	Test Statistic	df / Notes	Confidence Intervals / Bounds
One-sample variance $\sigma^2 = \sigma_0^2$	Normal population	$\chi^2 = \dfrac{(n-1)S^2}{\sigma_0^2}$	df $= n-1$. Exact under normality.	For σ^2: $\dfrac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}} < \sigma^2 < \dfrac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}}$ Lower: $\sigma^2 > \dfrac{(n-1)S^2}{\chi^2_{1-\alpha,n-1}}$ Upper: $\sigma^2 < \dfrac{(n-1)S^2}{\chi^2_{\alpha,n-1}}$
Two-sample variance $\sigma_1^2 = \sigma_2^2$	Two independent normal populations	$F = \dfrac{S_1^2}{S_2^2}$	df $= (n_1-1, n_2-1)$. Exact under normality.	For $\theta = \sigma_1^2/\sigma_2^2$: $\dfrac{S_1^2/S_2^2}{F_{1-\alpha/2,n_1-1,n_2-1}} < \theta < \dfrac{S_1^2/S_2^2}{F_{\alpha/2,n_1-1,n_2-1}}$ Lower: $\theta > \dfrac{S_1^2/S_2^2}{F_{1-\alpha,n_1-1,n_2-1}}$ Upper: $\theta < \dfrac{S_1^2/S_2^2}{F_{\alpha,n_1-1,n_2-1}}$